

Challenges and trends in automotive battery recycling

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Niklas Kronemeyer^{1,2}✉, Hannah Mittag³, Chris Gabrisch⁴, Jens Leker¹, Simon Lux^{1,3}, Richard Schmuch³ & Fabian Duffner⁴

The electrification of the automotive industry is driving a rapid global expansion of battery production, which generates substantial volumes of waste from both manufacturing scrap and end-of-life batteries. Consequently, recycling capacities must scale significantly from 2026 onwards, while recyclers face major challenges: meeting regulatory requirements, establishing efficient reverse logistics and coping with the growing complexity of return streams, system designs, cell formats, and evolving chemistries. Here we provide a comprehensive overview of current recycling technologies and assess their compatibility with both state-of-the-art and next-generation batteries, highlighting the technical and economic challenges of managing diverse battery waste. We show that although established processes are largely suitable for today's lithium-ion chemistries, future systems such as sodium-ion and solid-state batteries will require process adaptation. Beyond technical aspects, we identify seven key trends to enhance recycling economics, optimize material flows and support the development of a fully circular battery industry in the future.

The ongoing electrification of the automotive sector drives dynamic growth in the battery industry, creating rising demand for critical raw materials such as lithium, nickel, cobalt and copper¹. Their extraction entails substantial economic and environmental impacts, and uneven geographical distribution raises geopolitical risks². Recycling and reusing materials from waste batteries can enhance economic viability, improve sustainability and mitigate geopolitical tensions. Governments worldwide are therefore promoting battery recycling through regulations that require process optimization (for example, minimum recovery rates), secure demand for recycled material (for example, recycled content standards) and prevent landfill disposal (for example, extended producer responsibility)³. Consequently, all batteries must be recycled, and recycling is expected to ramp up alongside battery production.

Battery production is expected to follow three industrialization phases: a warm-up phase until -2025 for establishing and optimizing the first gigafactories; a scaling phase with widespread factory rollout;

and slow saturation from 2040 onwards⁴. Due to battery lifetimes, the recycling industry is expected to follow a similar trajectory with a time lag of around 10 years³, implying that significant scaling will be needed from 2030. Battery recyclers should prepare now, as securing market share during scaling requires proven production processes.

However, the recycling industry faces major challenges due to the high variance in returned batteries regarding cell technology, cell design, system design and origins, as well as the lack of standardized recycling processes⁶.

In this Review, we discuss the current status of automotive battery recycling, including technical and economic challenges, as well as emerging trends towards a circular economy. We analyse the technological feedstock complexity and evolving market landscape, showing that feedstock complexity is high and dynamically changing over time. We then assess the compatibility of current recycling processes with this feedstock complexity, highlighting that variations in battery chemistry in particular require adjustments in recycling processes.

¹Institute of Business Administration at the Department of Chemistry and Pharmacy, University of Münster, Münster, Germany. ²Porsche Consulting, Stuttgart, Germany. ³Fraunhofer Research Institution for Battery Cell Production FFB, Münster, Germany. ⁴Volkswagen, Salzgitter, Germany.

✉e-mail: niklaskronemeyer@gmail.com

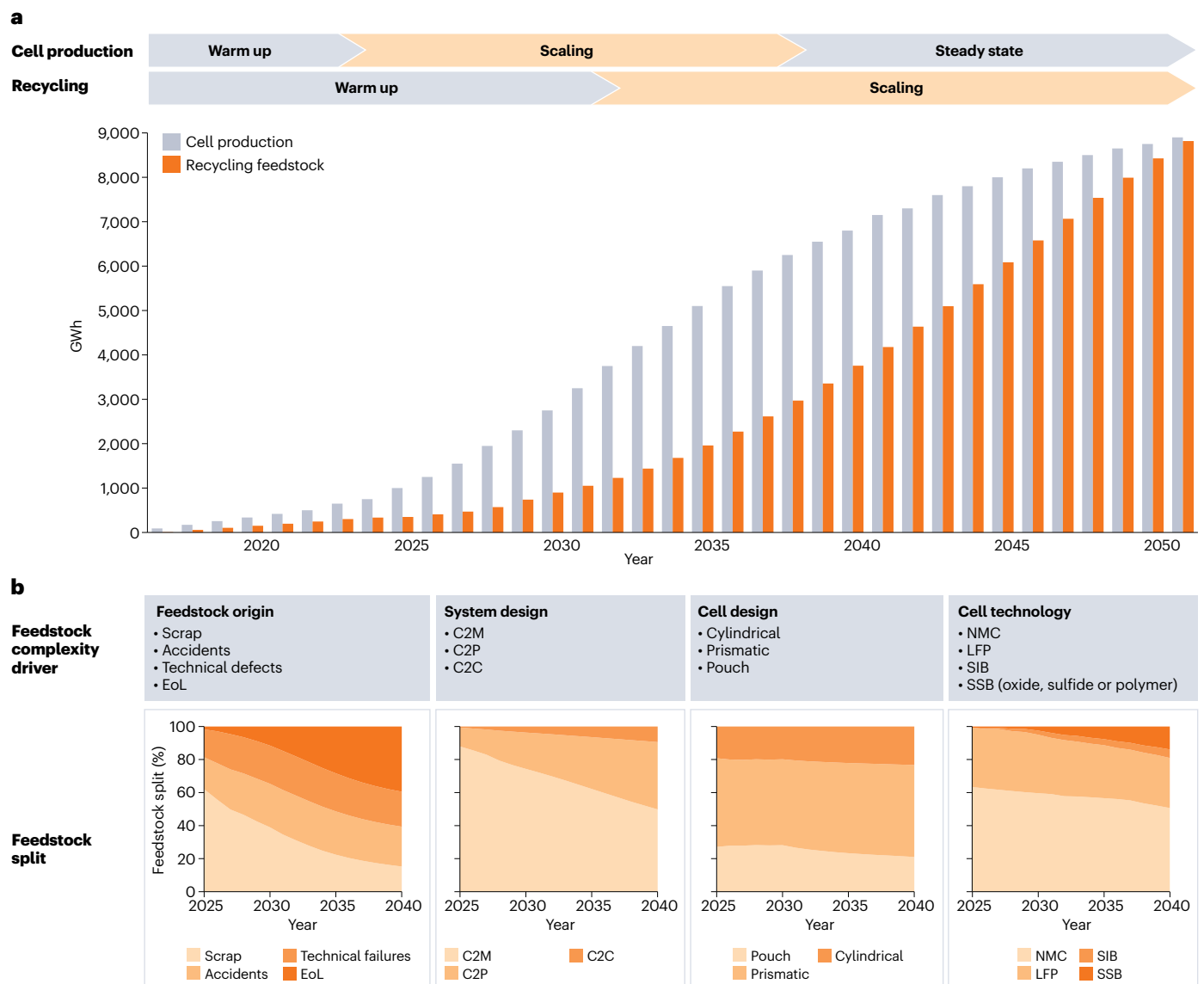


Fig. 1 | Battery cell production levels, recycling feedstock volumes, and recycling feedstock shares of the main feedstock complexity drivers.

a, Current and predicted battery cell production levels and recycling feedstock volumes until 2050. Battery production has transitioned from an initial warm-up phase into a rapid scale-up phase characterized by substantial capacity expansion across major markets. From 2035–2040 onwards, capacity growth is expected to moderate as markets approach higher levels of maturity. In contrast, recycling volumes remain in an early warm-up phase. A pronounced scale-up phase is expected from around 2030, driven by increasing EoL battery flows and tightening regulatory requirements. The battery cell production volume data

are from refs. 4,14,93. The return volumes are based on our own calculations. The data, premises and formulae for these calculations can be found in Source Data Fig. 1. **b**, The feedstock is characterized by a high and dynamic degree of complexity, driven in particular by four key complexity drivers: (1) the feedstock split of the battery origins production scrap, accidents, technical defects and EoL batteries; (2) the feedstock split of the battery system designs C2M, C2P and C2C; (3) the feedstock split of the cylindrical, prismatic and pouch battery cell designs; and (4) the feedstock split of the battery cell technologies (chemistries) NMC, LFP, SIB and SSB. The feedstock splits are based on our own calculations.

Finally, we examine the economic conditions of battery recycling and show that the growing share of nickel- and cobalt-free batteries poses economic challenges for current recycling systems.

Battery recycling feedstock complexity and market

Feedstock origin

Feedstock originates from production scrap, accidents, technical defects or end-of-life (EoL) vehicles⁷, with their relative contributions changing over time. As shown in Fig. 1, current return volumes are dominated by production scrap—accounting for 70% in 2025—due to high scrap rates during the ramp-up of new battery factories⁷. As the

battery electric vehicle fleet ages, the share of collected batteries from EoL vehicles and other non-scrap sources is expected to rise, reaching around 60% by 2030.

System design

System designs can be categorized into three concepts: cell-to-module (C2M), cell-to-pack (C2P)⁸ and cell-to-chassis (C2C) designs⁹. C2M, the state-of-the-art design, assembles cells into modules and modules into packs. In the C2P design, which represents the next generation and has already been industrialized by some companies, modules are omitted and cells are directly integrated into the pack, thereby reducing costs and saving space⁸. The C2C design goes further by integrating cells

directly into the vehicle chassis⁹, offering additional cost and volume advantages, but remains at a pre-industrial stage.

Cell design

Rechargeable automotive batteries use prismatic, cylindrical and pouch cell formats¹⁰, which differ in shape, size, capacity and casing material¹¹. Prismatic and pouch cells typically use aluminium housings, whereas cylindrical cells often use nickel-plated steel¹². Chemistries and cell formats can currently be freely combined, but cylindrical cells are rare in low-cost chemistries (for example, lithium iron phosphate (LFP) and sodium-ion batteries (SIBs)), which are split mainly between prismatic and pouch designs¹³. Market trends indicate a shift towards prismatic cells¹⁰. Nickel–manganese–cobalt oxide (NMC) cells are currently distributed across all three formats but are expected to see growing shares of prismatic and cylindrical designs¹⁰.

Cell technology

The chemistry of lithium-ion batteries (LIBs) is a key driver of feedstock complexity. Currently, the market is dominated by lithium NMC and LFP cathodes, both paired with graphite anodes¹⁴. NMC cells are costlier due to their critical raw material content but deliver superior performance, making them suitable for high-performance applications¹⁵. LFP cells serve cost-sensitive segments with lower performance needs but are rapidly gaining market share¹⁶. SIBs, which replace lithium with sodium, promise cost and sustainability advantages due to sodium's abundance¹⁷, although their higher ion mass and lower voltage result in reduced energy density¹⁷. SIBs are expected to penetrate low-cost electric vehicle segments and stationary storage¹⁸, with first products already commercialized¹⁹. Solid-state batteries (SSBs) use solid electrolytes, enabling high-energy dense lithium metal anodes and improved safety¹³. Currently, multiple concepts based on various solid electrolytes (namely polymers, sulfides or oxides, often in hybrid) exist^{13,20–23}, but challenges in interface stability, manufacturability, cost and cell design must be solved before large-scale adoption^{13,24,25}.

Battery recycling market

Driven by rapid battery market growth, the recycling sector is evolving dynamically, with both established chemical companies and start-ups developing innovative process technologies. Recycling processes can be classified into black mass production and pyrometallurgical, hydrometallurgical or direct recycling routes¹¹. Pyro- and hydrometallurgical processes recover individual materials, some of which can be reused to produce new cathode active materials (CAMs), whereas direct recycling yields the CAM as the final product²⁶.

The largest recycling capacities are in Asia, as production scrap dominates the feedstock, and battery production is concentrated in Asia. Additionally, black mass is currently being shipped from Europe to Asia for processing²⁷. Asian recyclers (for example, from China and South Korea) benefit from better feedstock access, economies of scale and local experience, creating high entry barriers for European and North American players²⁷. However, the ramp-up of local battery supply chains in Europe and North America is driving growth in regional recycling capacity announcements (Table 1). Hydrometallurgy remains the dominant extraction route globally, whereas direct recycling has not yet been industrialized. Overall, capacities are projected to expand significantly by 2030. As battery production increasingly localizes in Europe and the Americas, closer geographical proximity between cell manufacturing and recycling facilities can create operational synergies. These include more efficient logistics, improved transparency and alignment of material specifications, and better conditions for scaling up recycling operations.

Status and challenges of the recycling process

Battery recyclers must account for a dynamic market landscape and high feedstock complexity when scaling up industrial facilities. A wide

range of process technologies are available, broadly clustered into pyrometallurgical, hydrometallurgical and direct recycling routes²⁸. Each route comprises three process phases: (1) pre-treatment; (2) black mass production; and (3) material recovery with multiple process steps²⁹. However, depending on the process design, not all process steps are required. For example, black mass production is optional for the pyrometallurgical route, as batteries can be directly fed into a smelting process³⁰ (Fig. 2).

Return logistics

The return logistics of EoL batteries involve considerable effort and account for up to 40% of recycling costs³¹. Different points of return for EoL batteries (for example, scrap yards, workshops, dealers and automotive original equipment manufacturers (OEMs)) and battery scrap (for example, battery cell and system manufacturers) must be considered in the logistics concept. In addition, safety and regulatory requirements pose a challenge. Waste batteries are classified as hazardous goods and packaging requires short-circuit prevention³². Even stricter regulations apply to damaged and defective batteries, requiring special UN-certified containers, which are expensive due to the lack of a standardized shape, size and weight of battery packs³². The implementation of a hub-and-spoke logistics network promises cost benefits⁵. Here the batteries are collected locally and then transported to decentralized spokes for black mass production⁵. The black mass is then transferred to the centralized hubs for material recovery, which saves costs as safety requirements are lower than for EoL batteries³². Return logistics is currently at an early stage of development and therefore inefficient. However, cost potential can be achieved by exploiting economies of scale and synergies through standardization and optimized collection concepts⁶.

Pretreatment

First, the returned batteries are analysed for their condition and chemical composition. Theoretically, batteries with a good condition can be repurposed for a second-life application (for example, stationary energy storage)^{33,34}. However, returned batteries remain a black box regarding cell chemistry and state of health (SoH), as manufacturers protect their intellectual property and disclose little information, posing a major challenge for analysis. As manual or charge–discharge classification is time consuming and has low accuracy³³, analytics and sorting are often skipped in industry. The battery passport, a digital ID containing relevant value-chain information³⁵, is expected to facilitate analysis. However, due to long battery lifetimes, even by 2040 many batteries will return without a passport. Therefore, new cost- and time-efficient processes (for example, X-ray transmission or machine-learning tools) must be developed^{36,37}. In general, return logistics, analytics and sorting are associated with high safety requirements, as the residual energy of the batteries can cause fire and high-voltage accidents. To mitigate risks, after sorting, the remaining electrical energy is extracted through electrical, thermal or electrochemical processes for battery deactivation^{38,39}. Electrical discharging is labour intensive, as batteries are manually connected and deeply discharged, but energy can be reused⁴⁰. During thermal pyrolysis, which is typically carried out under vacuum conditions at 600 °C, the remaining energy content is reduced and organic components are evaporated, resulting in exhaust gas⁴¹. As no manual connection of batteries is required, this method promises high scalability. In the electrochemical discharge method, batteries are immersed in an aqueous salt solution (for example, Na₂CO₃, Na₂SO₄ or NaCl)³⁹ and electrolytic reactions drain the remaining energy from the batteries. This process is currently a niche application due to its duration (3–20 h), material losses in the solution^{39,41} and casing corrosion (for example, steel or aluminium), which leads to electrolyte leakage and contamination⁴². New approaches such as redox shuttles or non-electrochemical discharge using conductive powders (for example, flake graphite or copper powder) have been reported^{42,43}. Process know

Table 1 | Overview of selected battery recycling facilities in Europe and North America, including company, location and process classification based on official announcements

Region	Company	Location	Process	Annual capacity (kt _{battery})	
				2025	2030+
Europe	SungEel HighTech	Bátonyterenye, Hungary	Mechanical	35	
	PreZero	Meppen, Germany	Mechanical	30	60
	Battri	Saint-Laurent-Blangy, France	Mechanical	15	35
	BASF	Schwarzeide, Germany	Mechanical	15	
	Librec	Biberist, Switzerland	Mechanical	12	
	Fortum	Harjavalta, Finland	Hydrometallurgical	8	70
	Umicore	Hoboken, Belgium	Pyrometallurgical	7	
NAR	Redwood Materials	Sparks, USA	Mechanical and hydrometallurgical	60	
	Redwood Materials	Berkeley County, USA	Mechanical and hydrometallurgical	50	
	Cirba Solutions	Lancaster, USA	Mechanical	50	90
	Ascend Elements	Covington, USA	Mechanical and hydrometallurgical	30	
	American Battery Technology Company	Reno, USA	Mechanical and hydrometallurgical	20	100

For announced black mass capacities, a conversion factor of 0.4 kt black mass per kt battery capacity was assumed. The data were taken from the literature and are summarized in a source data file. BASF, Badische Anilin- & Soda-Fabrik.

how during deactivation is important for the subsequent steps, as the state of charge can affect the recycling efficiencies^{11,44,45}.

After deactivation, the battery is dismantled into modules or cells by removing the housing, battery management system and cooling system. Thorough removal of adhesive and foam residues is essential to prevent downstream contamination. Owing to design complexity, varied dimensions and fasteners and the use of structural adhesives, disassembly is typically manual or semi-automated, making it time consuming and costly⁸. In addition, specially qualified workers are required due to high voltage risks¹¹. Accurate disassembly is necessary to fulfil the recovery requirements of the EU Battery Regulation of 80% of the total mass by 2030⁴⁶. Standardization of battery systems and design for recycling in battery development would enable automated dismantling¹¹, resulting in a time saving of 85% compared with manual dismantling⁸. Design-for-recycling measures include the reduction of screws, welded parts and structural adhesives using tapes, clips, wire bonding or debondable adhesives^{47,48}. However, mechanical stability, thermal conductivity and electrical insulation of the battery must be ensured⁸.

Black mass production

After disassembly, the black mass, which contains the critical materials (for example, lithium, nickel and cobalt), is produced (Fig. 3) by removing inactive metals (for example, aluminium, iron, copper and steel), plastics and the liquid electrolyte. Black mass production is a crucial process phase, as the purity and consistency of black mass are vital for efficient recovery of the critical metals in battery recycling⁴². First, the batteries are crushed either dry or wet with liquids (for example, water)^{36,42}. The violent friction of the materials during mechanical shredding or comminution causes heating and temperatures of up to approximately 300 °C⁴⁹. To prevent explosions and the formation of toxic gases due to unwanted reaction of the liquid electrolyte with ambient air, shredding is carried out under an inert atmosphere (for example, nitrogen or argon) or under vacuum conditions^{42,50}. A combination of different equipment (for example, two-blade rotor crusher, impact crusher, hammer mills or planetary ball mills), sieves and rotation speeds can be used to minimize the particle size of the resulting powder^{36,42}.

After shredding and comminution, the liquid electrolyte is removed and the black mass is dried using thermal (infrared or convection), vacuum, mechanical (centrifugation) or chemical methods (solvent or supercritical CO₂ extraction)^{11,28,50}. Since residual moisture can contaminate downstream processes, its removal is critical; consequently, the Chinese government has limited the maximum moisture content to 2%⁵¹. The liquid electrolyte is recovered via condensers, either in direct liquid (Fig. 3g–i) or gaseous form⁵² (Fig. 3d–f). Shredding is often performed under vacuum, enabling concurrent electrolyte recovery. The collected electrolyte solution contains organic solvents (linear and cyclic carbonates), lithium salt (LiPF₆), additives (for example, fire retardants), binders and impurities formed by reactions between electrolyte and electrode⁵⁰. The electrolyte collection method determines the impurity of the electrolyte compound. Although chemical processes promise high purity of the recovered electrolyte⁵⁰, thermal processes are also suitable for simultaneous removal of the binder⁴². For reuse of the electrolyte in battery production, the complex impurities of the collected electrolyte compound caused by chemical reactions between electrolyte and electrodes must be removed⁵⁰. Multiple processes such as filtration, active carbon decolorization, molecular sieve purification, distillation or exchange resin deacidification can be conducted, but achieving battery-grade quality of regenerated electrolytes is still challenging and associated with high costs⁵⁰.

After electrolyte collection and drying, the impurities of the black mass are removed⁵². Typical steps are sieving, magnetic separation and density separation³⁶. During sieving, the coarser particles and metal foil fractions can be separated⁴² and eventually fed back for further comminution. Magnetic separation removes the magnetic particles (for example, iron) and density separation removes the light particles (for example, plastics and graphite). The sequence and iteration of individual process steps are tailored by recyclers depending on feed-stock and process design⁵³.

Material recovery

The resulting black mass is either directly recycled or further treated via pyrometallurgical or hydrometallurgical routes to recover valuable metals. Currently, recycling primarily targets cobalt and nickel for

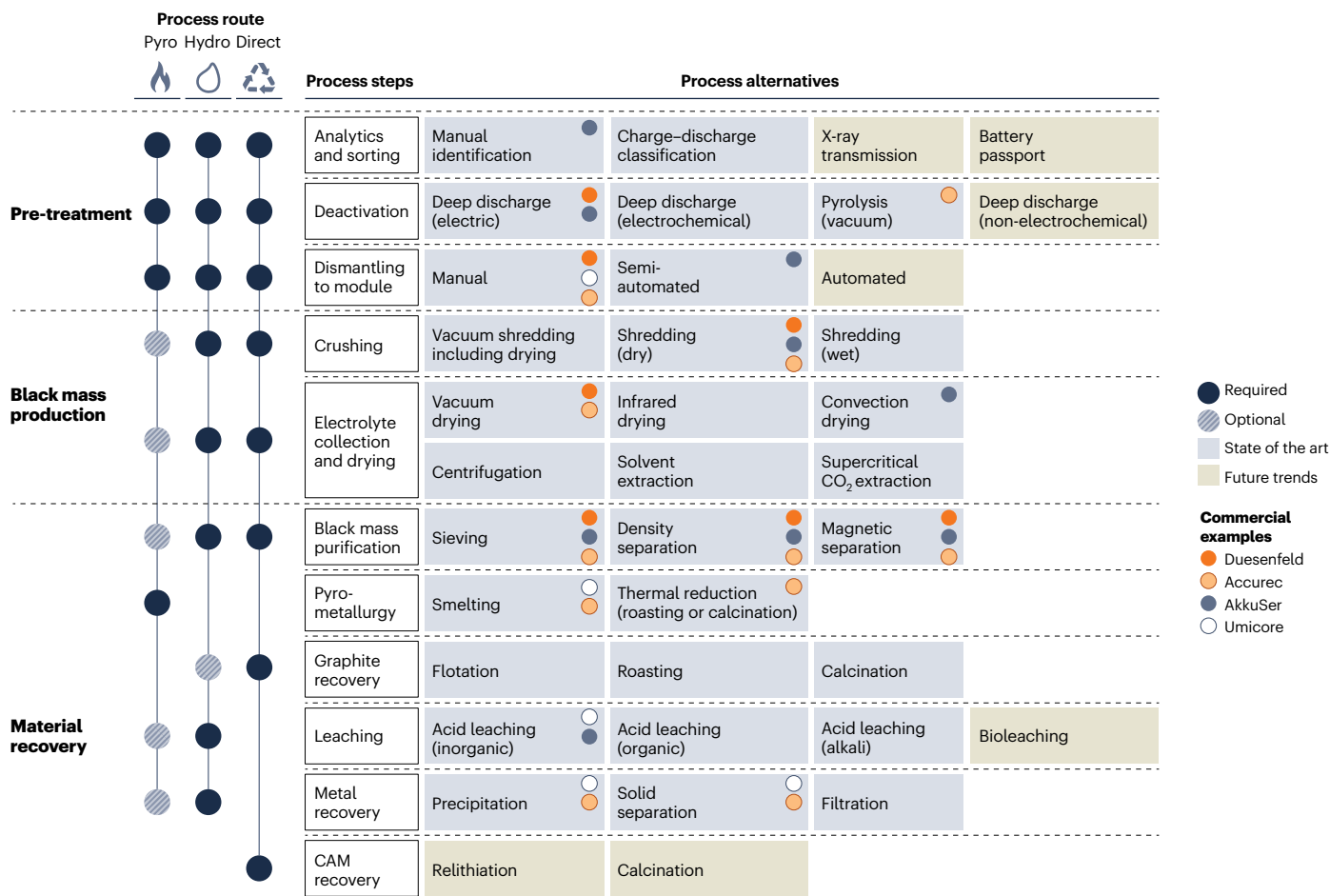


Fig. 2 | Routes and steps of the battery recycling process, with suggested alternative technologies. The classification of process routes was derived from the existing literature^{11,53,94}, but process routes at industrial and pilot scales are highly specialized. Consequently, several process steps are considered

optional within these routes, and numerous alternatives—both state-of-the-art and emerging options—are available for these specific steps. The exemplary commercial process routes were derived from the literature^{11,53,94}. Hydro, hydrometallurgical; Pyro, pyrometallurgical.

economic reasons, whereas lower-value materials such as graphite are rarely recovered⁵⁴. Regarding the market shift towards low-cost batteries (for example, LFP), regulations become important to promote the recycling of less economically attractive materials⁵⁵.

Pyrometallurgy is a well-established and robust recycling method (for example, by Umicore¹¹), in which high-temperature treatments such as roasting or smelting are used to produce a mixed metal alloy and slag from deactivated batteries or black mass⁵⁵. Although the metals from the alloy can be further extracted using hydrometallurgical processes, recovery of lithium and aluminium oxides from slag is economically challenging⁵⁶. The organic components and graphite are incinerated during the process, serving as an energy source and reducing agents⁵⁷. Since the materials partly contain fluorine, expensive gas treatment is necessary⁵⁶.

If no pyrometallurgical process was conducted, graphite can be recovered using selective flotation or filtration^{20,54}. Although some flotation processes are already in commercial use, the purity of recycled graphite is insufficient for battery production⁵⁴. Higher purity can be achieved by initially leaching the black mass before filtration⁵⁴. Despite the lower economic value of graphite compared with nickel or cobalt, graphite recovery has gained increasing interest since the biggest graphite producer, China (71% market share of mining in 2023⁵⁸), announced graphite export restrictions^{2,59}.

The metals from the black mass or alloy can then be further separated using hydrometallurgy, which utilizes aqueous solvents to extract and recover precursors in battery-grade quality^{11,20}. Hydrometallurgical

processes could be very sensitive to variation in feedstock composition and typically involve leaching, precipitation and solvent extraction steps⁶⁰. During leaching, metals of interest (for example, nickel, cobalt and manganese) are brought into solution⁶⁰. Several inorganic (for example, H₂SO₄ or HCl) or organic acid leaching agents (for example, citric or ascorbic acid) are reported, which are associated with high environmental impact and cost^{52,61}. Bioleaching based on bacteria or fungi, a more sustainable but slow alternative, is being researched^{49,62–64}. After leaching, the metal hydroxides are precipitated (for example, by adding NaOH)⁵¹. Whereas nickel, manganese and cobalt hydroxides are separated and dried, lithium remains in solution and can be recovered through further precipitation and electrodialysis steps^{60,65}. The resulting precursors can then be reintegrated into the production of new CAMs^{20,60,66}.

In contrast with hydrometallurgy, direct recycling retains the crystal structure and morphology of the CAM, allowing the active material to be directly reintegrated into battery production^{29,66}. However, aged CAMs must be upgraded (for example, using re-lithiation) to compensate for potential lithium losses, which are typically higher in nickel-rich chemistries (up to 13%)²⁶. The idea of upcycling goes one step further by converting an outdated CAM stoichiometry into a state-of-the-art framework^{56,67}.

Opportunities and challenges of direct recycling

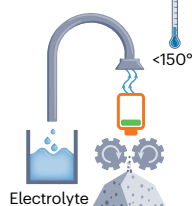
Direct recycling has not yet been industrialized, but the development of the process is of great interest⁶⁶. In contrast with pyrometallurgical

Crushing**a Dry shredding**

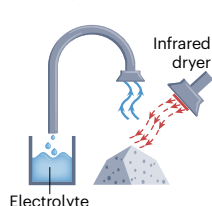
Nitrogen atmosphere

**b Wet shredding**Argon atmosphere
Solution (for example, water)**c Vacuum shredding**

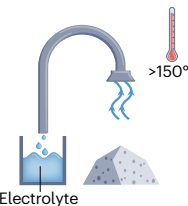
Vacuum

**Electrolyte collection and drying****d Infrared drying**

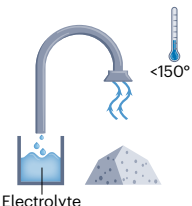
Nitrogen atmosphere

**e Convection drying**

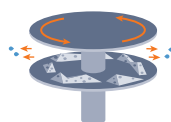
Nitrogen atmosphere

**f Vacuum drying**

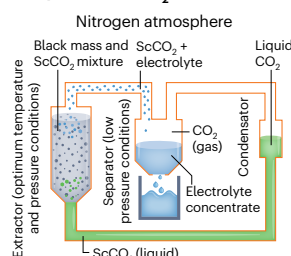
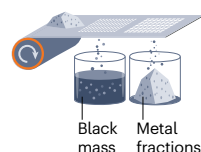
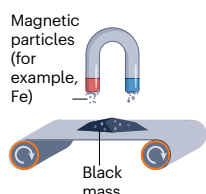
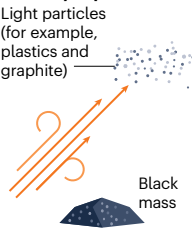
Vacuum

**g Centrifugation**

Nitrogen atmosphere

**h Solvent extraction**

Nitrogen atmosphere

**i Supercritical CO₂ extraction****Black mass purification****j Sieving****k Magnetic separation****l Density separation****Fig. 3 | Schematic describing the process of the black mass production phase for lithium-ion batteries.**

a–c, First the batteries are shredded. This process can take place either dry in a nitrogen atmosphere (**a**), wet in an argon atmosphere (**b**) or in a vacuum (**c**). The challenge is to achieve the finest possible particles. **d–i**, After shredding, the liquid electrolyte is recovered and the powder (black mass) is dried. Drying and electrolyte recovery can be carried out thermally (via infrared (**d**) or convection drying (**e**)), under vacuum (**f**), mechanically (**g**) or chemically (via solvent (**h**) or supercritical CO₂ extraction (**i**)). To achieve the highest possible purity of the black mass, these processes can also be combined with each other. **j–l**, The black mass is then refined and impurities are removed. Coarser particles can be removed by sieving (**j**). Magnetic (**k**) and density separation (**l**) are used to remove magnetic particles (for example, iron (Fe)) and light particles (for example, plastics). To achieve a high level of purity, all three processes are often used in succession.

recycling is of particular interest due to the higher share of CAM production in the total costs and the lower value of the individual raw materials (for example, lithium, iron or phosphorus)²⁹. However, there are three main challenges in the industrialization of the process.

First, the process and the resulting electrochemical performance of recycled CAMs are sensitive to impurities or chemical compositions. Therefore, very thorough pre-sorting of the input stream is necessary, which presents a considerable technical challenge³⁰. Additionally, increased requirements apply to the purity of the black mass, as impurities (for example, residual aluminium or binder) lead to poor electrochemical performance¹¹.

Second, cycled batteries contain mechanically degraded CAMs, due to mechanical and electrochemical stress during cycling. This is especially the case for polycrystalline layered-oxide CAMs⁶⁹. The degree of degradation is individual and increases with the service life of a battery. Reconstruction of the particle morphology of degraded CAMs without calcination is technically challenging to not possible²⁶.

Third, non-destructive direct recycling recovers specific CAMs while preserving their stoichiometry⁶⁶, meaning that only the original cathode materials can be recovered. Considering the high dynamic of battery technology development, the CAM stoichiometry of the EoL batteries might be outdated at the time of its recovery³⁰. Although several studies investigate upcycling^{70,71}, the technology remains at an early stage and its feasibility is yet to be proven²⁶.

Therefore, direct recycling is likely to be limited to production scrap, which offers simple collection and sorting and has a low degradation level^{56,66}. Electrode scrap is even more preferred, as no contact with electrolyte and no charge–discharge cycle has been conducted yet. Additionally, the recovered CAM is state of the art and can be fed back into battery production.

If the technical feasibility of direct recycling could be validated, this process offers an opportunity for in-house recycling for cell manufacturers and direct integration into battery production. This will be particularly interesting in managing the mandatory recycled material content in Europe⁷².

Environmental legacy issues

In addition to processual considerations, environmental legacy issues are becoming increasingly relevant for battery recycling. A particular concern is the handling of solid and liquid waste streams containing persistent and potentially hazardous substances, especially per- and polyfluoroalkyl substances (PFASs), which are recognized as persistent environmental contaminants⁷³. In current LIBs, polyvinylidene fluoride is used as a binder (approximately 3.75 g kg⁻¹; ref.⁷⁴), contributing to PFAS loads in EoL batteries.

Recent studies have identified bis-perfluoroalkyl sulfonimides—a PFAS subclass used in lithium-ion batteries—as highly mobile and toxic compounds^{73,75}. They have been detected in landfill leachates and surface waters near battery disposal sites⁷⁶, raising concerns over long-term groundwater contamination. Although the European Union has initiated regulatory steps to phase out PFAS substances used in the production of binding agents (for example, polyvinylidene fluoride and polytetrafluoroethylene) necessary for batteries⁷⁷, these rules apply only regionally. Moreover, legacy batteries and imports from non-EU regions will continue to introduce PFASs into recycling streams for more than a decade.

Recyclers must adapt to emerging PFAS regulations and legacy contamination risks. Effective monitoring and treatment of PFAS-containing sewage will be essential to ensure long-term environmental compliance. Initial findings suggest that pyrometallurgical processes operating at temperatures of up to 1,600 °C can effectively mineralize PFASs, whereas lower-temperature hydrometallurgical routes (<600 °C) may lead to incomplete degradation and potential PFAS release⁷⁵. However, in high-temperature processes, complete exhaust air treatment is necessary due to the hazardous gases produced

or hydrometallurgical processes, direct recycling recovers the CAM instead of separate materials, which promises considerable economic advantages⁶, as energy-intensive CAM production can be omitted (Fig. 4)⁶⁸. Thus, the revenue potential for recyclers doubles for NMC batteries and quadruples for LFP batteries⁶⁸. For LFP batteries, direct

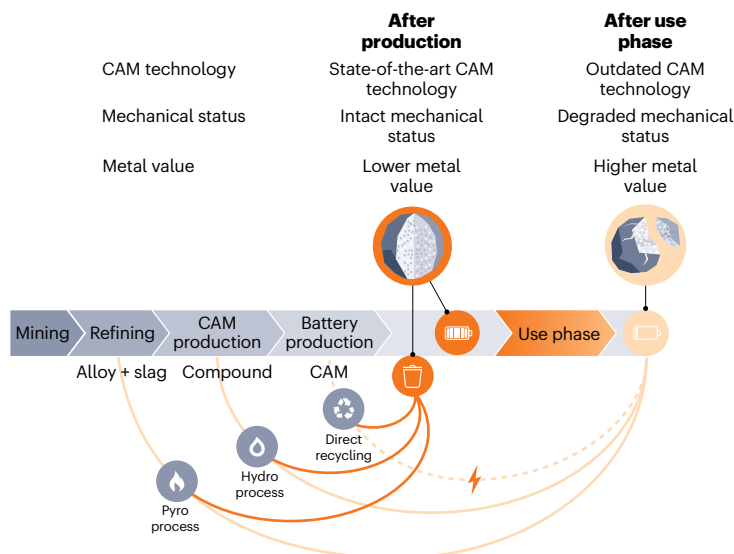


Fig. 4 | Routes for a circular economy of batteries. Ideally, the recovered materials from the pyrometallurgical, hydrometallurgical and direct recycling process routes are reintegrated into the battery supply chain. As each recycling process generates a different process output, the recovered materials are fed back into the supply chain at different stages. Generally, the battery supply chain includes mining and refining and production of the CAM, which is then used

in battery production. After reaching EoL, the CAM is typically mechanically degraded and technologically outdated, which makes direct recycling approaches challenging. Moreover, the potentially higher metal value content of outdated CAM compositions compared with state-of-the-art materials often makes elemental material recovery economically more attractive than direct recycling.

(including hydrofluoric acid)⁷⁵. In general, further research is needed to prevent the environmental emissions produced through recycling.

Process compatibility with feedstock complexity

Most recycling studies examine specific battery chemistries (NMC and LFP) from economic or environmental perspectives, without considering battery origin, system design or cell design^{68,78}. As recyclers face increasingly complex and dynamic feedstocks, assessing process compatibility with this complexity is critical to maintain competitiveness during scale-up. Here we analyse key feedstock complexity drivers and evaluate their impact on recycling process compatibility (Fig. 5).

Battery origin

Beyond logistics, the origin of batteries strongly influences their preparation for subsequent processing steps. Batteries with technical defects or those from accidents pose particular safety risks due to their high voltage and therefore require strict safety standards, specialized precautions and extensive employee training to prevent short circuits, fires and injuries. Although state-of-the-art processes can be applied, contactless diagnostic methods (for example, X-ray transmission⁷⁹ or battery passport scanning) and deactivation techniques (for example, deep discharge in solution or powder, or pyrolysis) provide important advantages for defective batteries or those involved in an accident. Once deactivated, such batteries can be processed similarly to conventional EoL batteries, although accident-induced mechanical deformations complicate automated dismantling. In contrast, production scrap batteries generally have a well-defined chemical composition and condition, except for defective scrap from formation, where the state of charge is unknown. For most production scrap, analysis can be omitted and depending on whether the scrap is electrode, cell or system level, disassembly may be simplified or unnecessary. In these cases, the battery origin no longer plays a decisive role for subsequent processing.

System design

The system design also influences the preparation phase. The shift from C2M to C2P architectures increases disassembly complexity, as part

counts rise from 12 modules in the Renault Zoe (C2M) to 178 cells in the BYD Han (C2P)⁸. However, if C2M battery systems are also dismantled to cell level, the C2P battery system shows clear advantages, as the number of parts to be removed is significantly lower⁸. C2C systems allow a further reduction of parts^{8,9} but require new processes during the preparation phase. So far, there are few data on C2C systems, but besides providing energy they fulfil safety and stability functions in the chassis. The dimensions as well as materials and stability of housing differ from those of C2M and C2P systems⁹, resulting in different handling requirements within the preparation phase and new requirements in deactivation and dismantling. Thus, new processes must be developed for C2C systems.

Cell design

Cell design affects both dismantling and black mass purification, although current equipment can generally accommodate different formats. Cylindrical cells typically have lower energy content than pouch or prismatic cells, leading to higher removal counts during dismantling. Their shape also creates cavities that are often filled with adhesives (for example, glue or epoxy), which complicates dismantling and contradicts with EU Battery Regulation targets for improved reparability^{11,46}. Prismatic cells, in contrast, can hold significantly higher energy, creating elevated safety risks during degassing¹¹. Differences in housing materials (for example, nickel-plated steel versus aluminium foil) may require further adjustments in black mass purification to achieve targeted granularity and purity. Whereas aluminium housings from pouch and prismatic cells can be reused, viable applications for nickel-plated steel from cylindrical cells remain undeveloped.

Cell technology

Cell technology has the strongest impact on recycling processes. For LFP, black mass production requirements largely mirror those of NMC⁸⁰, but worker toxicity risks after shredding are significantly lower (1.0×10^{-7} comparative toxic units for humans per kilogram of material (CTUh kg⁻¹) for LFP versus 17.0×10^{-7} CTUh kg⁻¹ for NMC)⁸¹. With LFP, the recovery steps for nickel and cobalt are omitted. Iron can be recovered

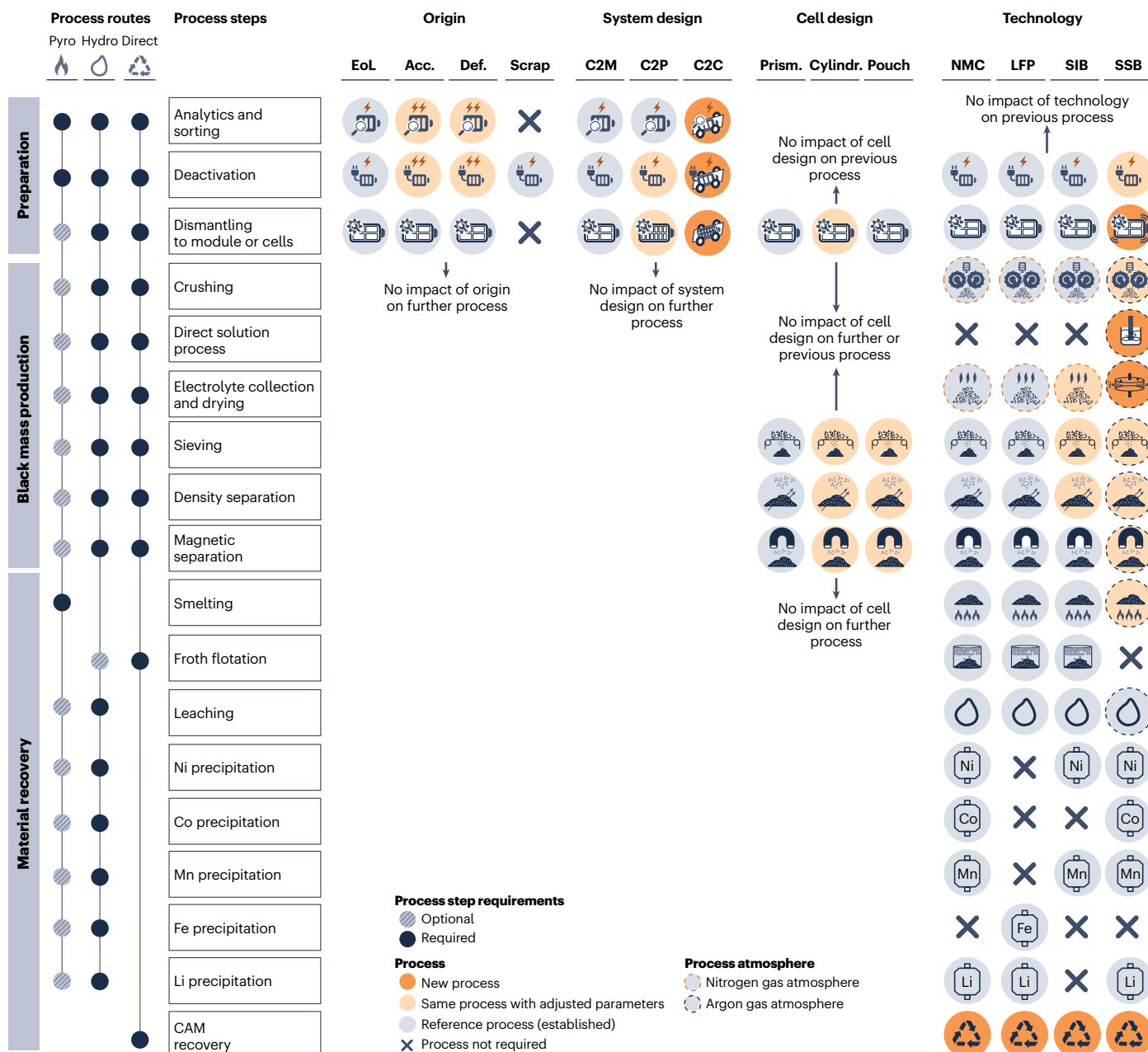


Fig. 5 | Impact of battery feedstock complexity drivers on recycling process requirements and compatibility with reference process. Battery origin has impact on the preparation phase. Batteries from accidents or technical failure must be processed under increased safety requirements, due to high voltage risks. If battery scrap is conducted some process steps could be omitted. The battery system design also influences the preparation phase. Since no modules and potentially other bonding methods are used, deactivation and dismantling in C2P differs from C2M. Current processes are probably not suitable for the preparation of C2C systems and new processes must be developed. The cell designs have an influence on dismantling, as cylindrical cells in particular cause cavities due to their design, which are often filled with adhesives. In addition, the housing materials of the cell designs differ, which must be considered in black

mass production. The battery chemistry has the greatest influence. The process requirements of LFP are the same as those for NMC batteries, but different materials have to be recovered during material recovery. For SIBs, process adjustments are necessary in black mass production due to different electrolytes and current collector foils, as well as in material recovery due to different metals. Substantial adjustments are necessary for SSBs. Due to the SSBs specific cell and system designs, adaptation of the preparation phase is necessary. The recovery of the solid electrolytes requires new processes (for example, direct solution process) and the metallic lithium anode leads to challenges in black mass production regarding equipment maintenance and inert gas atmosphere. Acc., accidents; Co, copper; Cylindr., cylindrical; Def., defects; Li, lithium; Mn, manganese; Ni, nickel; Prism., prismatic.

either individually (the destructive approach) or as FePO_4 without damaging the crystal structure (the non-destructive approach)^{80,82}. Conventional leaching agents (for example, H_2SO_4) enable non-selective recovery, whereas additives such as $\text{NH}_3\text{H}_2\text{O} + \text{NaOH}$ and Na_2CO_3 allow lithium and iron separation¹⁶. Although the destructive approach remains state of the art, the non-destructive route requires less leaching agent and offers economic and environmental benefits⁸⁰.

SIBs follow the same preparation process as LIBs, but complete deactivation is essential, as sodium metal and sodiated carbon materials are more reactive than lithium or lithiated graphite¹⁸. Reported SIB chemistries show generally lower toxicity risks than LFP (0.1×10^{-7} – 0.7×10^{-7} CTUh kg^{-1})⁸¹, although vanadium-containing CAMs can even exceed NMC toxicity levels (54.0×10^{-7} CTUh kg^{-1}) and Prussian-blue chemistries may release highly toxic cyanogen⁸³.

Black mass production requires minor process adjustments due to different electrolytes (for example, NaPF_6)¹⁸, but purification is simplified since both electrodes use aluminium foils¹⁷. Furthermore, the material recovery differs substantially compared with that from NMC. A variety of cathode materials (for example, polyanionic or Prussian-blue based) are reported and besides replacing lithium with sodium, the majority do not use nickel and cobalt¹⁷. Additionally, hard carbon instead of graphite is used as the anode material¹⁸. Thus, multiple recovery processes are not required, but new processes could be implemented (for example, hard carbon or sodium)¹⁸. However, sodium recovery remains economically unattractive due to its abundance and low value¹⁸.

In contrast, SSBs require fundamental process changes driven by: (1) their specific cell and system designs; (2) the metallic lithium anode; and (3) the solid electrolytes.

Most SSB cell concepts show substantial volume change during operation¹³. To maintain continuous contact between solid layers¹³, specific cell and system designs applying high stack pressures are required^{24,25}. Although industrially viable designs are not yet established, dismantling processes will need to be adapted accordingly, including measures to protect workers from the mechanical stress involved. A second major challenge is the metallic lithium anode. Due to its high reactivity (for example, with water and nitrogen from the ambient air⁸⁴), the cells must be processed in an argon atmosphere after opening until the lithium has been removed^{20,22}. Other chemistries also require inert atmospheres, but only until electrolyte collection⁵², and a cheaper nitrogen atmosphere is usually sufficient¹¹. Additionally, the adhesive property of metallic lithium⁸⁴ poses a challenge for black mass production and requires regular equipment maintenance. To reduce the area of protection gas (for example, argon) and simplify black mass production, early-stage lithium recovery after shredding (for example, by thermal treatment or supercritical CO_2 extraction) is discussed in the literature^{20,85}, but the industrial feasibility is still to be tested⁸⁵. Furthermore, graphite recovery is no longer required if lithium metal anodes are considered.

Due to the solid electrolytes, the battery cells are more robust against physical stress⁸⁶ and the electrolyte is more difficult to remove^{20,87}. The shredding process can be carried out with standard equipment, but must be adapted to the increased robustness²³. In addition, the different mechanical properties of the solid electrolytes¹³ must be considered. Although the mechanical crushing of brittle oxidic electrolytes is easy, soft sulfidic electrolytes can cause problems²⁰. New processes are required to recover the electrolyte due to its solidity, and the requirements for the processes differ for each type of solid electrolyte (polymer, sulfidic or oxidic). In polymer-based solid electrolytes (for example, those based on poly(ethylene oxide)), the polymers and lithium salts (for example, lithium bis(trifluoromethanesulfonyl)imide) can be dissolved by soaking in water in combination with mechanical energy, and Li_2CO_3 or Li_3PO_4 can be recovered from the lithium salts using conventional methods (for example, hydrometallurgy)²⁰. Sulfidic solid electrolytes can be recovered by dissolution in solvents (for example, *N*-methylformamide, methanol or ethanol), followed by precipitation²⁰. For oxidic solid electrolyte recovery, thermal treatments with a temperatures close to the sintering temperature (up to 1,050 °C)³⁸ have been reported. In the case of sulfidic solid electrolytes, high protective measures are necessary, as toxic H_2S can be formed. However, H_2S formation can be prevented by adding metal oxides (for example, Fe_2O_3)³⁸.

Overall, it can be observed that joint recycling of LIBs (NMC and LFP) and SIBs is technically feasible but increases the risk of cross-contamination and decreases the recovery rate for individual materials⁵⁴. Therefore, separate recycling of NMC, LFP and SIBs is recommended for economic competitiveness. SSB batteries require substantial process adjustments and would cause considerable technical and safety problems if added to state-of-the-art recycling facilities.

Economics and future trends towards a circular economy

The economics of battery recycling remain challenging, as recent reviews combining techno-economic modelling and machine-learning approaches show that, under current cost structures, total process costs often exceed the value of recovered materials—especially for nickel- and cobalt-free chemistries such as LFP and SIB (Fig. 6a)⁶. Although nickel- and cobalt-rich NMC batteries can yield positive gross margins under favourable metal-price conditions, the average profitability across real-world mixed feedstock remains marginal⁸⁸.

Recycling is regulatorily mandated in all major markets, transmitting economic pressure across the battery supply chain and embedding recycling within a policy-driven circular economy that shapes design, production and EoL pathways⁴⁶. Despite its central role in enabling circularity, battery recycling remains at an early industrial stage, with inefficiencies and underutilized capacity. Major opportunities exist to improve economics through targeted levers across the value chain (Fig. 6b).

Design for recycling

Battery pack and cell design strongly influence recycling cost structures⁸. As discussed in the section process compatibility with feedstock complexity, increased structural integration in C2P and emerging C2C designs leads to higher disassembly complexity and costs compared those of with C2M systems. Standardized fasteners, modular pack architectures and debondable adhesives can reduce disassembly efforts by up to 80% compared with state-of-the-art manual disassembly⁸. Design for recycling also improves safety, reduces pre-treatment costs and limits cross-contamination in black mass, which directly enhances downstream yields⁴⁷. As regulatory obligations on recyclability are internalized, design for recycling is becoming an important differentiator in next-generation battery platforms.

Generating new revenues through second-life applications

Battery cycle stability has improved substantially, such that automotive batteries often retain significant functional value (SoH > 80%) at the end of vehicle life³³. Preserving this residual value through second-life use (for example, stationary storage or low-demand mobility) supports circularity, although accurate and standardized SoH assessment remains a key challenge⁸⁹. Although second-life deployment does not directly raise recycling margins, it improves system economics by lowering the cost per cycle, increasing EoL predictability and enabling closer OEM–recycler integration, with standardized diagnostics reducing analytical costs at final recovery⁸⁹.

Standardized return logistics

Return logistics accounts for up to 40% of total recycling costs, driven by hazardous-goods regulations and fragmented return points³². Standardized pack interfaces, safety protocols and container formats, combined with regional collection hubs, enable shorter transport distances for high-risk batteries and allow long-distance transport to be shifted to lower-risk black mass¹¹.

Process optimization

Process optimization is the largest direct lever for improving recycling profitability. Semi-automated dismantling of standardized packs can cut labour by 50–80%⁸, and improvements in shredding, electrolyte removal and drying increase black mass purity and recovery efficiency⁵⁰. Optimized hydrometallurgical processes, including tailored leaching, selective precipitation and solvent extraction, further reduce operating expenditure^{52,60}. Specializing in homogeneous feedstock streams (for example, separate NMC, LFP or SIB lines) enhances yields, lowers purification costs and enables scale efficiencies across the value chain⁵².

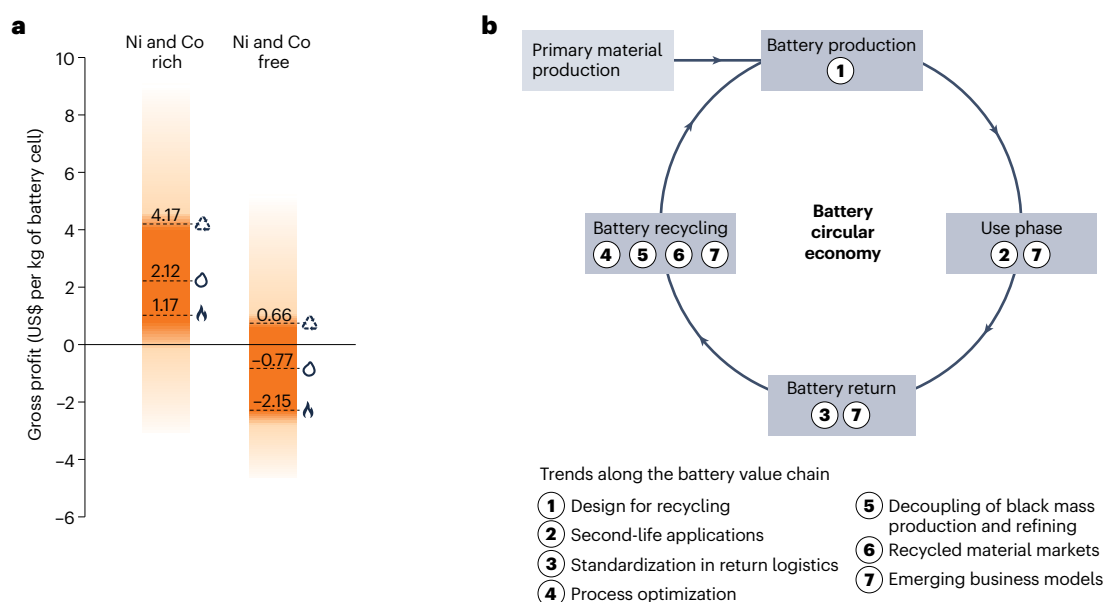


Fig. 6 | Economics of battery recycling and trends along the battery value chain. **a**, Gross profit of nickel- and cobalt-rich versus nickel- and cobalt-free batteries, defined as the total revenue from recovered materials minus the processing costs. The values were derived from a machine-learning-based literature review on battery recycling economics⁶. Note that industrial data will probably differ from academic cost modelling. The dark shaded areas represent average values across different recycling routes, with the dashed lines representing the average reported value by process cluster (that is, pyrometallurgical, hydrometallurgical or direct). The light shaded areas indicate reported outliers. The wide spread reflects differing assumptions on logistics

and energy costs, material prices, laboratory expenses and recovery efficiencies in the literature. The reported gross profit strongly depends on nickel and cobalt content, with the recycling of nickel- and cobalt-free chemistries remaining economically challenging. Overall, direct recycling shows the highest profitability, followed by hydrometallurgical and pyrometallurgical processes. **b**, To improve recycling economics—particularly for nickel- and cobalt-free chemistries—and to enable a circular battery economy, seven key future trends are identified. These trends span the entire battery value chain (that is, battery production, the use phase, battery return, EoL processing and material reintegration).

Decoupling black mass production and material recovery

A structural decoupling of upstream mechanical processing and downstream refining is emerging globally⁹⁰. Upstream steps (that is, collection, deactivation, dismantling and shredding) are logistics driven and benefit from regional infrastructure, whereas downstream hydrometallurgy and pyrometallurgy require scale and chemical expertise²⁹. Decoupling enables specialization and improved cost structures while black-mass trading-market transparency increases, spurred on by China's introduction of classification and quality standards for traded black mass⁵¹. Such standards help to ensure traceability, reduce uncertainty for refiners and enable price differentiation by composition and purity. Additionally, logistics costs can be reduced by 20%⁵ by implementing hub-and-spoke systems, where decentralized black-mass production feeds centralized refining⁹⁰.

Recycled material markets

Recycling revenues are currently dominated by a few high-value streams (lithium, nickel, cobalt, aluminium and copper)⁵⁶. Although reintegrating these materials into battery production reduces primary demand, regulations (for example, the EU Battery Regulation) increasingly mandate minimum recycling efficiencies across all fractions, including low-value streams such as graphite, iron phosphate and electrolytes⁴⁶. For LFP, sodium-ion and other emerging chemistries, profitability depends on monetizing these high-mass, low-value materials. Graphite can be purified and reused to battery-grade or industrial standards⁵⁹; recovered iron phosphate has potential as a precursor for fertilizers or catalysts¹⁶; and electrolyte recovery, currently associated with high costs and technological challenges, is gaining attention due to both potential reuse and regulatory pressure to mitigate PFAS-related environmental risks⁷⁵. Developing downstream markets for secondary materials is essential as battery chemistries shift to lower-metal-value systems, otherwise recycling revenues remain narrow and vulnerable to metal-price volatility.

Business models

Battery recycling has traditionally relied on tolling or material ownership, exposing recyclers and OEMs to price volatility, fragmented feedstock flows and limited risk sharing⁸⁸. Emerging business models embed recycling more strategically into the battery value chain.

Closed-loop models, including in-house recycling by OEMs, integrate recycling into circular business strategies, providing guaranteed feedstock, enabling optimized utilization of recycling equipment and securing materials for OEMs⁹¹. In-house recycling is particularly relevant for the direct recycling of production scrap, which will accumulate in large volumes during the ramp-up of new battery facilities⁶⁶. It allows OEMs to capture additional value from recovered materials and maintain control over quality and process optimization, strengthening circularity while reducing reliance on third-party recyclers. Additionally, closed-loop strategies can include second-life applications, where used batteries are redeployed in low-performance applications such as stationary energy storage systems before final material recovery, capturing additional value across the product life cycle³³.

Producer responsibility organizations (PROs) are established to ensure compliance with regulatory requirements throughout the battery take-back and material recovery process⁴⁶. They manage logistics, collection and compliance, taking on responsibility for steering and coordinating EoL flows until recovered materials re-enter production. By standardizing processes, aligning incentives across OEMs, recyclers and logistics providers and leveraging scale advantages, PROs reduce coordination costs, mitigate regulatory risk and limit system fragmentation⁹².

Together, these models transform recycling from a standalone service into a strategically embedded function, enabling high-capacity utilization, lower financing risk and more stable economics across the battery supply chain.

Conclusions

The recycling industry must prepare for significant scaling, as feedstock volumes are expected to triple by 2030 and increase further towards 2040. This growth will be accompanied by rising feedstock complexity, creating challenges for both industrialization and business models. The industrialization pathway can be divided into two phases. In the next decade, recycling will be dominated by state-of-the-art system designs (C2M) and chemistries (NMC and LFP). Processes and equipment for pyrometallurgical and hydrometallurgical recycling are already established, with only limited technical hurdles remaining. The primary challenge lies in economic optimization—reducing operating expenditure and capital expenditure while maximizing recovery efficiencies—which can be addressed through specialization in homogeneous feedstock streams.

Beyond 2030, the feedstock mix will shift as the share of production scrap decreases and new systems (C2P and C2C) and technologies (SIB and SSB) enter the market. Direct recycling offers economic potential since no resynthesis of CAMs is required, but its applicability is probably restricted to production scrap, as degraded cathode material and heterogeneous EoL batteries pose major technical barriers. With the scrap volume share decreasing year by year, the long-term market potential of direct recycling appears limited. At the same time, C2P and C2C architectures will complicate analytics, sorting, deactivation and dismantling while new chemistries create further challenges. Integrating SIBs into existing processes may only marginally reduce recovery rates and utilization, but SSBs—with lithium metal anodes, adhesive designs and solid electrolytes—will require fundamentally new process development. Moreover, as cell chemistries and material demands continue to evolve, recyclers must regularly reassess which outputs remain relevant; for example, large-scale adoption of SSBs could render the recycling of liquid electrolytes or graphite obsolete, undermining incentives for developing efficient processes for these materials.

These technological and feedstock challenges directly shape the economics of battery recycling, which remain particularly tight for low-metal-value chemistries such as LFP and sodium-ion chemistries. Regulatory mandates embed recycling within a policy-driven circular economy, but profitability depends on targeted levers, including design-for-recycling, second-life deployment, standardized logistics, process optimization and decoupled upstream–downstream operations. Revenues still rely on high-value materials, yet emerging markets for graphite, iron phosphate and electrolytes are essential. Strategically embedded business models, including closed-loop approaches and PROs, stabilize feedstock, optimize utilization and align incentives across the value chain, enabling a scalable, circular battery industry.

To capture market share in the scale-up phase beyond 2030, recyclers must use the current warm-up period to refine processes, reduce costs and establish viable business models. Above all, early and close integration with automotive OEMs will be critical to secure feedstock volumes, anticipate evolving technology roadmaps and remain competitive in an increasingly complex recycling landscape.

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N.K., H.M. and R.S. conceived of the article idea and wrote the manuscript. C.G., S.L., J.L. and F.D. reviewed the manuscript. All authors edited the manuscript.

Competing interests

N.K. is an employee at Porsche Consulting. F.D. and C.G. are employees at Volkswagen. The remaining authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Niklas Kronemeyer.

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